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Metadata as report and support

A case for distinguishing expected from fielded metadata

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Steps

- ⇒ Two ways of looking at metadata
  - → Metadata as **reporting** about data, information to the data user
  - → Metadata as **supporting** work with data, specifically the work of the data publisher
- ⇒ Example
  - → Comparing expected metadata with fielded metadata (processing)
- ⇒ Questions

Background: VarInfo

- ⇒ A prototype for managing metadata, used at SIDOS
  - → www.sidos.ch/mmg/vi/html/toc.htm
- ⇒ Concepts further developed for the MetaDater poject, yet not integrated in final model

Reporting

I - The ‘reporting’ perspective

- ⇒ Metadata as a report on data construction...
  - → Meaning (wordings)
  - → Representativity (collection method)
  - → Relevance (indexes)
  - → Intention (concepts and hypotheses)
- ⇒ ... published to meet the needs of data users
  - → Publication: One dataset with the matching metadata
- ⇒ Characteristics or those metadata
  - → Static – final state, even if successive versions
  - → Selective – only published data are documented
  - → ‘Passive’ – They don’t work for you, they do just describe data

Once upon a time...the life cycle stance

- ⇒ Need for a simplification of the presentation of the DDI model, which grows more and more complex
- ⇒ Observation: all metadata are not needed at every stage of the data definition, collection, processing and analysis processes
- ⇒ Response is: to split up the model into modules
  - → Study, data collection, logical product, physical data product, physical instance, archive...)
  - → Phase in process and/or levels of information

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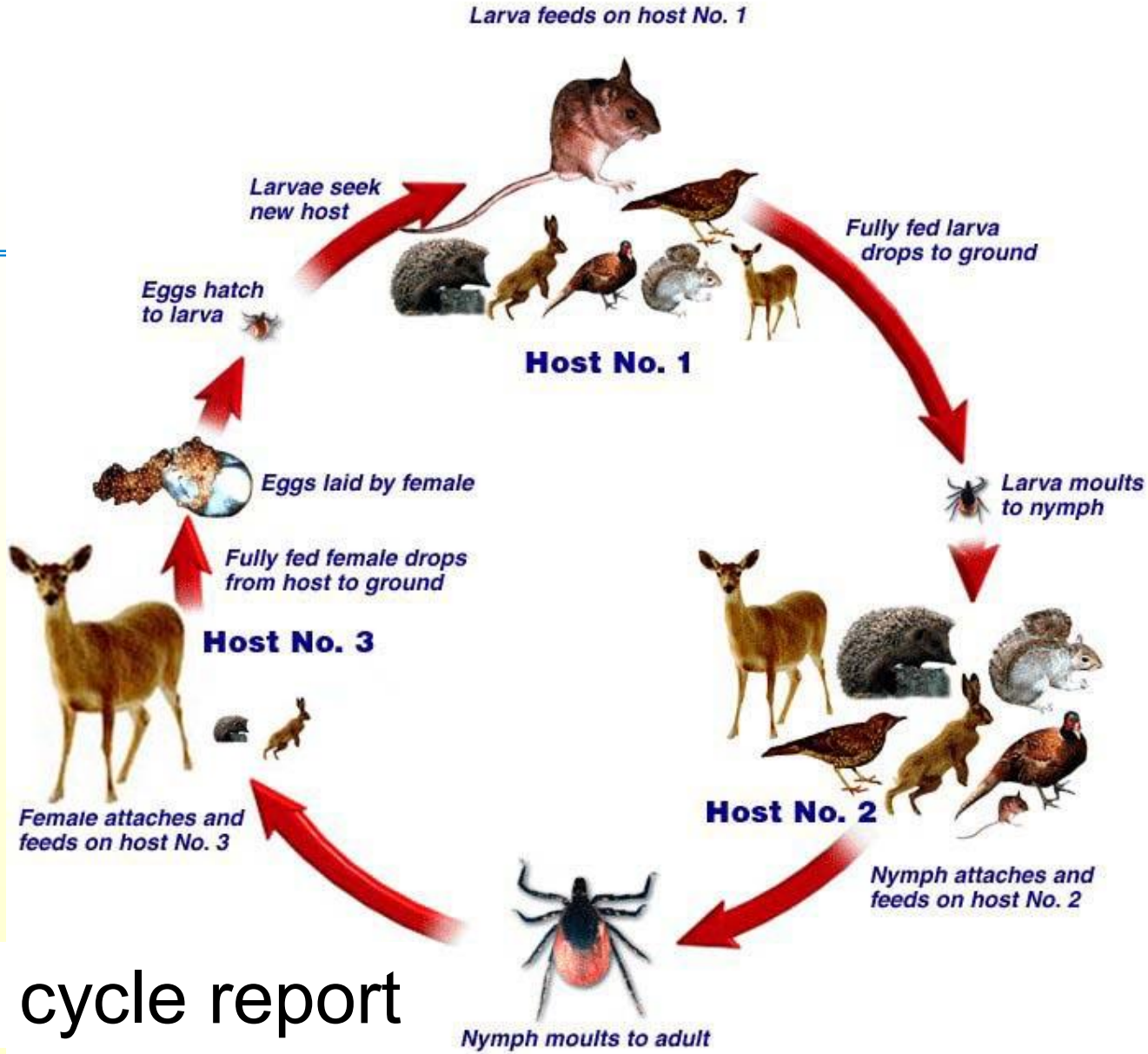
Life cycle report



The relative size of the animals approximates their significance as hosts for the different tick life cycle stages in a typical woodland habitat.

Courtesy of Dr Jeremy Gray and Bernard Kaye

The life cycle report: take a questionnaire

- ⇒ Modalities of the report
  - → Printout of the questionnaire
  - → File (PDF or text editor)
  - → Oject in the DDI 3 ‘data collection module’
- ⇒ Variables appear as part of an other object
  - → Data definition file (classical)
  - → Logical Data Product module in DDI 3
- ⇒ Questions and variables can be linked
  - → Textual reference or electronic
  - → The link is **descriptive**
  - → Questions belong to a questionnaire, variables to a data file

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Life cycle support

II – The supporting perspective

- ⇒ The supporting perspective supposes a life cycle approach
  - → No support is needed for a fixed object (data/metadata as to be published)
  - → **Support**: various activities must be supported over time
  - → **Action**: There is a ‘before’ and an ‘after’
- ⇒ It is a cycle of actions, not only a cycle of states
  - → **Use cases**: you need a description of the action to get the model, which will really support that action

Excursus: Behind the ‘support’ idea, a system

- ⇒ Documenting means reporting on something
  - → Only needed : a format (e.g. DDI 2)
- ⇒ Supporting work means having a system capable of action
  - → Store (database)
  - → Procedures (application)
  - → A data model including elements to control procedures
  - → ... various states of the data and metadata (not only versions!)
  - → A process model, defining the steps to be gone

Rescuing endangered metadata (a use case)

- ⇒ Data publishers (archives) often get metadata and data in a poorly coordinated way
  - → Some version of a printed questionnaire
  - → A data file the primary researcher worked with (constructions, recodes, badly documented variables)
- ⇒ Primary researchers may get from the data collector a data file which does not match the questionnaire
  - → Variations in variable names , codes, variables lists
- ⇒ Both need a consistent data / metadata set
  - → Matching information with a pencil and paper method may be very time-consuming and leaves nothing to be of any further use

Introducing: Expected metadata The Q/V

- **Questions imply a variable definition**
  - you ask a question to get a specific kind of measure. The basic metadata unit is not just a question, but a **question & variables element**
  - Those variable definitions have the status of **expectations**
- The link between a question and the expected variables is an **organic**, not a casual one. Q and expected V's belong together
- The link between the fielded and the expected variables (and hence the questions) is **to be assessed**
  - Consistent variable names?
  - All expected variables present?
  - Are there additional fielded variables?
- The link between a question and the fielded variables is **composed of an organic and an assessed part**

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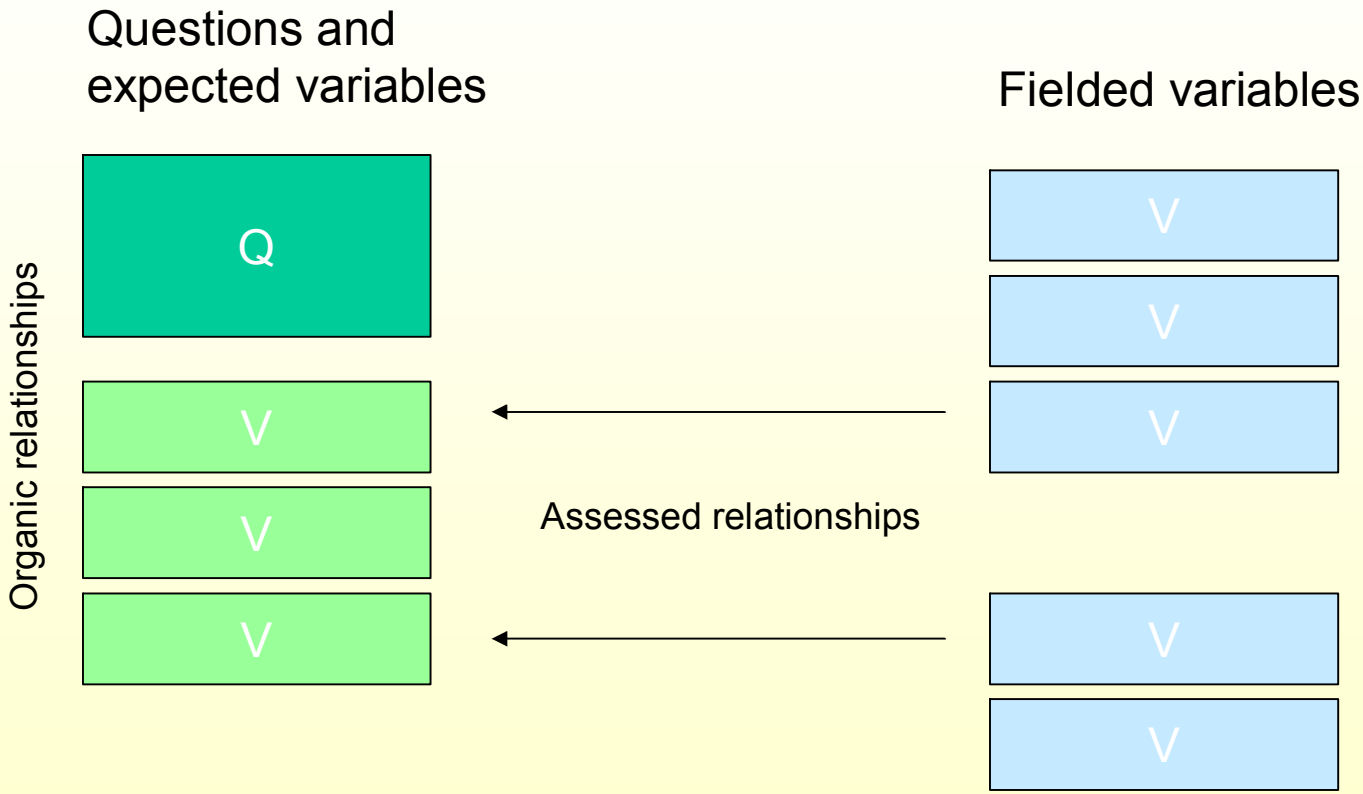
The schema



Data processing use case: the setting

- ⇒ Given:
  - → System, Study, Questions & expected variables
  - → A semi-documented data file of the SPSS kind, coming from the field
- ⇒ Metadata construct:
  - → **Two distinct stores** for variable level metadata
    - **Expected metadata**, expressed as a question and response categories or another kind of variable definition
    - **Fielded metadata**, expressed as a file definition
  - → **Tables establishing correspondence** between expected and actual metadata, where a mismatch occurs
    - Establishe mediated match
    - Define correction

Data processing: the procedures

- ⇒ Identify mismatches
  - → Variable names (lists of non-matching names)
  - → Values of coded variables: lists of non-matching codes; example: list of values in a data file, which are not defined in the variable definition as expected
- ⇒ Correct mismatches
  - → Variable names
  - → Values of coded variables
- ⇒ Run corrections
  - → Procedure depends on the data store used
  - → SPSS files: the program computes and executes a syntax file

- ⇒ Sometimes, it is the expectations, which have to be amended...
- ⇒ The same information is used for
  - → correction (supporting)
  - → documentation of the correction (reporting)
- ⇒ There is no additional reporting work to do (‘documentation’)
  - → Just process, the process will leave a trace (‘documentation’)

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Expected metadata: Answer categories directly related to variable labels

- ⇒ The Q/V concept integrates answer categories (questions) and variable labels (variable definitions)
  - → **Functionally equivalent**
  - → Only difference: **length**, because of limited store for labels
- ⇒ Answer categories and expected labels:
  - → Answer categories should be the labels if they don't exceed the allowed length
  - → Either lets store all short versions, and long versions only if necessary
  - → ...or store answer categories of any lenght, and additional short versions if the answer category is too long
- ⇒ Possible action: label any data file with expected labels (instead of « correcting the file »)

Closing questions

- ⇒ Shall we stay with reporting metadata, or add supporting metadata?
- ⇒ Which use cases are central enough?
- ⇒ Can we, as a **small** community, manage the way from the format to the system?
- ⇒ Which organisation, which funding?

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Next generation support